Unit 8 Economic performance

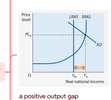
8.1 Economic growth and the economic cycle

- difference between short-run and long-run economic growth
  - economic growth: an increase in productive potential or the rate at which output of all goods and services produced by and economy is increasing.
    - Short-run economic growth is the percentage change in the total output of an economy from one time period to the next
    - Long-run economic growth is the average annual percentage change in the productive capacity of an economy over a number of years
  - Elustrating the difference between short-run and long-run economic growth
    -

      short-run economic growth
      - output of the economy has increased
        - more factors of production are employed
        - being used more efficiently
      - increase aggregate demand — employ more factor of production — to increase the output to meet the increase in demand
      - short-run economic growth is limited by the proudctive capacity of the economy
    -

      long-run economic growth
      - improvements in technoloy — supply-side improvements — productive capacity has increased
    -

      both short-run and long-run economic growth
      - if supply-side improvements lead to long-run growth but the demand does not change — firms will not increase production — spare capacity — point A in the diagram
- demand-side and supply-side determinants of economic growth
  - demand-side determinants of short-run economic growth
    - depends on the rate of growth of aggregate demand
    - aggregate demand increase — increase output — growth in real GDP — results from increase in component of aggregate demand — increase in investment and consumption — multiplier effect — increase in aggregate demand and output
    - short-run economic growth continue until the economy approaches to its normal capacity level of output — get close to normal capacity of output — cause price level to rise and will not have much effect on national output — will not lead to sustainable economic growth
    - economic growth means incomes are rising — increase in income spent on imports — deterioration in the balance of payments — government has to reduce aggregate demand — export-led growth: where a significant part of the increase in a country's real GDP results from exporting goods and services abroad — injection has multiplier effect and increase national output
    - aggregate demand falls — negative short-run economic growth — firms reduce production to stop the stocks of unsold goods increasing — real GDP will fall.
  - supply-Side Determinants of long run trend rate of Economic Growth
    - determined by changes in the supply-side performance
    - firms invest in new capital — spending on research and development — techonological change — firms train their workers — increase productivity and contributes to long-run growth
    - government spends money on infrastructure — new transport links improve efficiency — spending on education and training — improves skills, adaptability and productivity of workers — government funding of research and development
    - aggregate demand can also have the effect — aggregate demand is rising
      - firms are likely to make more profit and need to invest new capital — accelerator theory — firms are likely to spend more on reesearch and development — A steady, stable growth of aggregate demand is likely to encourage firms to take actions that improve the supply-side performance of the economy — vice versa
      - if aggregate demand is too high — unsustainable — inflation and balance of payments problems — reduce investment — damage the supply-side performance of the economy
- main phases of economic growth
  - economic cycle: the periods of expansion and contraction in economic activity around an economy's long-run trend rate of economic growth.
    - boom
      - Produce more than its normal capacity level of output
      - AD is very high, unemployment is low, inflation is high and rising, firms are making more profits
      - wages increase rapidly — difficult to recruit wokers
      - Shortage of labour and other factors of production
      - inflation — balance of payment deficit
    - recession
      - AD and economic activity are falling
      - Unemployment rises as output falls
      - Wage increases more slowly or may fall.
      - disinflation
      - firms make less profit
    - depression
      - Firms and households are pessimistic about the future
      - low AD and economic activity
      - unemployment is high
      - wages are low and may be cut as many firms are struggle to survive
      - deflation
    - recovery
      - Economic activity and real GDP start to increase
      - A rise in AD, firms increase output
      - unemployment falls
      - Inflation start to increase
      - Firms become more confident, investment will increase, productive capacity increases
    -

      The economic cycle
      - productive capacity of the economy is increasing
  - Economic indicators used to identify the phases of the economic cycle
    - real GDP
      - recession — real GDP is falling
      - depression — GDP may be stable — but well below the normal level of capacity of the economy
      - recovery — GDP is increasing — continue into boom phase
      - boom — growth of real GDP is limited to long-run trend rate of growth of the economy
    - unemployment
      - lagging indicator: a statistic shtat comes after an event has taken place
      - time lag: a period of time between one event and another
    - inflation
      - boom — rate of inflation increase — demand-pull inflation
      - recession — disinflation
      - depression — deflation
      - is a lagging indicator
    - investment
      - most volatile component of AD
      - recession or depression — cut back on investment — as AD and profit are falling
      - boom — profits are rising — find it difficult to meet the demand for their products — invest to increase productive capacity
- difference between positive and negative output gaps
  - Output gap = equilibrium level of national income minus the normal capacity level of output — =Ye-YN — output gap can be either positive or negative
    - 

      a positive output gap
      - occurs in the boom phase — accelerating inflation — deterioration in balance of payments
      - 解释图表 — YN,Ye是什么 — positive output gap=Ye-YN — rising inflation — reduce aggregate demand — fall in the level of national income — policies to control inflation
      - why — resources are being overuse
    -

      a negative output gap
      - occurs when the economy is in recession or a depression — the size will fall in the recovery phase of the economy cycle — lasts for a long time — disinflation or perhaps deflation
      - 解释图表 — YN,Ye是什么 — if persists — there will be depression — long period of high unemployment
        - Keynesian economists — government uses policies to increase aggregate demand
        - neoclassical economists — market forces will automatically create incentives
      - why — resources are underused
- how demand-side and supply-side shocks affect economic activity
  - How a demand-side shock within the domestic economy can cause economic activity to fluctuate
    - lose confidence — household will reduce spending and firms cut the investment — multiplier effect and accelerator effect — firms do not need to invest to increase their productive capacity — an economy that was growing steadily will then go into a recession
    - rising aggregate demand — firms invest to replace capital that has worn out — multiplier effect accelerator effect — rapid increse in aggregated demand — boom
    - reasons other than consumer confidence
      - Changes in interest rate
      - Changes in taxation and government spending
      - availability and cost of credit
  - How a demand-side shock outside the domestic economy can cause economic activity to fluctuate
    - change in exchange rate
    - change in global income
    - exports are a large part of aggregate demand
  - How supply-side shocks cause fluctuations in economic activity
    - price of oil increases — consuming countries have to spend more on imports — withdrawl — vice versa — increase in aggregate demand
    -

      a rise in the world market price of oil reduces economic activity in oil-consuming countries
      - cost of producing many products increases — shift SRAS curve to the left — movement along AD curve — real national income falls from Y1 to Y2 — economic activity has fallen — recession
      - cost-push inflation — fall in economic ectivity — in the long-run, reducing demand-pull inflation
      - firms cut output — increase in unemployment
      - impact on producers is different — oil is inelastic — export revenue increases — injection — increase aggregate demand
      - impact of changes in oil prices on consuming countries is usually much greater than the impact of changes in the prices of other commodities
  - The Kondratieff Wave
  - Fluctuations in economic activity are not only caused by economic shocks
  - 不考 了解一下就行

8.2 Employment and unemployment

- Mearing unemployment
  - counting people who claiming unemployment benefits
  - survey
  - international labour organisation(ILO)
- cyclical, structural, frictional and seasonal unemployment
  - cyclical unemployment: when the economy goes into recession and people cannot find work because aggregate demand is too low(demand-deficit unemployment or Keynesian unemployment)
    - AD falls
      - reduce output — firms do not need as many workers
      - profits also falls — reduce the number of people they employ — reduce costs of production
    - occurs when there is a negative output gap
      -

        reducing cyclical unemployment
        - economy is producing below its normal capacity level of output — cyclical unemployment
        - measures are taken to increase aggregate demand to AD2 — real national income will increase to YN — more jobs will be created — cyclical unemployment will fall
  - structural unemployment: long-term unemployment that occurs when the skills and location of the unemployed workers do not match the jobs available
    - often caused by lack of suitable skills for jobs available
    - causes
      - changes in techonology
        - lead to more efficient methods of production — labour productivity increases — fewer workers will be needed unless demand for product increases
        - skills required may be different — unable or unwilling to adapt these changes
        - development of new products — demand for other proudcts to fall
      - changes in the pattern of demand
        - demand for a product falls — people will be made unemployed — may be unable or unwilling to learn the skills needed
        - may be located in a different part of the country — people may find it hard to move to a different region
      - foreign competition
        - competitiveness of different countries — pattern of comparative advantage change
        - People employed have particular skills and the jobs are concentrated in particular regions
        - decline in a localised industry — affect other firms in the region — downward multiplier effect — many job losses in the local area — unwilling or unable to move to jobs in other regions
      - exhaustion of a natural resource — eg.: mining industries
    - exist because labour is not perfectly mobile
  - frictional unemployment: shor-term unemployment when people are between jobs
    - Economies are always changing — factors of production need to be reallocated
    - change jobs for many reasons and may be out of work for a short period of time
    - does not last long
  - seasonal unemployment: when people are unemployed at particular times of the year — Seasonally adjusted unemployment — removes the regular short-term seasonal variations from the figures
- distinction between unemployment and underemployment
  - underemployment: when a person is working but is like to work more hours, or is overqualified for his or her job and has skills that are not fully used — Aggregate demand does not fall as much as would happen if people were unemployed — government will not have to spend as much on welfare benefits
  - economic activity falls — unemployment and underemployments also increases
  - in recovery — people are underemployed are able to find a full-time job
- how employment, unemployment and underemployment are affected by both demand side and supply-side factors
  - productive capacity of the economy is influenced by the size of the labour force and labour productivity
  - level of national output is affected by changes in productivity, employment, and underemployment
  - Demand-side influences on employment, unemployment and underemployment
    - increase in aggregate demand — encourage firms to increase output to meet the demand — firms need more workers — firms may ask people who were underemployed to work more hours — people find it easier to find a job — make good use of their skills
    - aggregate demand starts to fall — it is expensive to train people and recruit new employees — firms will not reduce the number of workers in the short run — unless they believe AD will keep low — number of hours people work may be reduced
    - fall in aggregate demand — involuntary unemployment
  - supply-side causes
    - a high replacement ratio — A high replacement ratio means people are not much better off when they are working than when they are unemployed — If taxes on income and welfare benefits are high — reduce incentive to work — voluntary unemployment
    - immobility of labour
      - occpational unemployment
        - do not have transferable skills
        - there are not many opportunities to retrain and learn new skills
      - geographically immobile
        - have friends and family who live close by
        - Poor transport and communication links
        - difficult for people to find somewhere to live that they can afford
      - Government policy to help improve labour mobility
        - Pay for the retraining of works
        - Improve the transport and communication system
        - Provide affordable housing in regions where there are lots of job vacancies
    - technological change — increase structural and frictional unemployment
    - minimum wage laws: the lowest wage that can legally be paid to workers
      -

        - national minimum wage is set at Wmin — At a wage of Wmin — there would be involuntary unemployment equal to (N2 – N1) workers — number of jobs lost depends on the elasticity of demand for labour — demand for labour is very inelastic — fall in employment will be small
      - The effect of a minimum wage on employment and unemployment
      - higher minimum wage may not lead to job losses if there is an increase in labour productivity
    - trade unions — egotiate with employers to agree the wage paid to particular groups of workers — improve labour productivity
- global influences on employment and unemployment
  - world economic growth — increased exports — increase employment in export industries — multiplier effect — create jobs in other industries — vice versa
  - competitiveness
    - design and quality of products — ability to deliver on time — Productivity
    - exchange rate — fall in exchange rate — domestic prducts more competitive
    - affect net exports(X-M) and aggregate demand — increase in competitiveness — exports will increase — creates jobs and reduce unemployment
  - restriction on international trade
    - increase in tariffs and other restrictions — reduce exports — increase unemployment
    - might protect jobs in the country — but may lead to retaliation and trade war — fall in exports — unemployment is likely to increase in the industries affected

8.3 Inflation and deflation

- inflation, deflation and disinflation
  - high inflation can be very harmful for an economy
    - Keeping inflation low is an important objective of government economic policy
  - inflation: a general rise in price level
    - creeping inflation: when there is low rate of inflation that continues for many years
      - even low rate of inflation can be harmful
        - affects distribution of income
        - exports become uncompetitive — rate of inflation is higher than in other countries — create uncertainty& reduce economic growth& higher unemployment
      - low and predictable rate of inflation does not necessarily have these effects and can have some benefits
    - hyperinflation: when the rate of inflation is very high
      - is always harmful — value of money falls rapidly — economic activity falls significantly — why control the rate of inflation is the most important macroeconomic policy objectives in some countries
  - deflation: a fall in the general price level
    - malign deflation — bad deflation that occurs in a slump and is caused by low aggregate demand — reduce spending — lack of confidence — people believe goods will be cheaper in the future
    - Benign deflation — good deflation that is caused by improvements in technology, efficiency and productivity — delay spending
  - disinflation: when inflation exists but the rate of inflation is falling
    - In a recession, disinflation is much more common than deflation — often caused by a slower rate of increase in aggregate demand — not always by a fall in aggregate demand
    - Deflationary policies — often result in disinflation but do not often cause deflation
    - Reflationary policies — often result in a rise in the rate of inflation but they rarely cause hyperinflation
- demand-pull and cost-push influences on the price level
  - Demand-pull inflation: when the rise in the price level is caused by an increase in aggregate demand
    - most likely to occur when an economy is close to full capacity
    - if there is spare capacity — rate of inflation is likely to stay low
    - demand-pull inflationary process
      -

      - multiplier effect
      - it is difficult to reduce it without an increase in unemployment
    - demand-pull inflation
      -

      - main effect is on the price level
      - demand-pull inflation will continue and may accelerate — aggregate demand continues to increase more rapidly than the economy's underlying rate of economicgrowth
    - inflation is always caused by the money supply increasing too quickly — the monetarist view
  - Cost-push inflation: when the rise in the price level is caused by an increase in the costs of production
    - cost-push inflationary process
      -

    - cause
      - Rising labour costs perhaps due to an increased minimum wage
      - Higher global prices for components and raw materials
      - causes a rise in import prices — A depreciation in the external value of the exchange rate
      - An increase in indirect taxes such as higher VAT or environmental taxes such as a carbon tax
      - central bank has little control over the factors that cause it
    - cost-push inflation
      -

      - SRAS curve shifts to the left — price level rises from PL1 to PL2 — contraction in aggregate demand — movement along the AD curve — decrease in output and real national income falls from Y1 to Y2
      - but inflation cannot continue unless there is a sustained increase in aggregate demand — increase in costs can cause inflation in the short run
- how external events may affect domestic inflation
  - rise in commodity prices can start a cost-push inflation in consuming countries
  - fast growth of the global economy will cause a demand-pull inflation in countries that depend on exports — increase in the demand for commodities
  - global economy slows down — demand for exports and commodity prices will fall — disinflation
- how changes in the exchange rate affect inflation
  - depreciation in the exchange rate is likely to increase inflationary pressures
    - make domestically products more competitive — increase aggregate demand — demand-pull inflation
    - rise in the price of imports — increase cost of production — cost-push inflation
    - some imported products are included in the CPI — rise in price of imports — increase inflation as measured by the CPI
  - appreciation of the exchange rate will reduce inflationary pressures

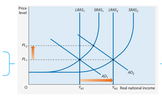
8.4 The balance of payments on current account

- current account of the balance of payments
  - BOP: a record of a country's financial transactions with rest of the world
  - trade in goods
    - Balance of trade= value of exports- value of imports
      - surplus — export>import
      - deficit — import>export
  - trade in services — banking, education, software
  - primary income
    - flow of income from residents of one country to another in return for investment and labour services
    - interest, dividends, rent, and wages
  - secondary income
    - money transferred from the residents of one country to another without any goods or services given in return.
    - government transfers (foreign aid) or private transfers
- meaning of deficits and surpluses on the current account
  - Current account balance: the sum of the balance of trade in goods and services + the primary income balance + the secondary income balance
    - example of the current account of the balance of payments

	12 billion	5 billion
Exports of goods	600	
Imports of goods	650	
Balance of trade in goods	-50	
Services balance	-200	
Balance of trade in goods and services		-150
Primary income balance		300
Secondary income balance		-50
Current account balance		20

- balance of payments account
  - 4 main sectors
    - the current account
    - the capital account
    - the financial account
    - net errors and omissions
  - Debit items: items that lead to an outflow of foreign currency. (import)
  - Credit items: items that lead to an inflow of foreign currency. (export)
- Factors that influence a country's current account balance
  - affected by the competitiveness of domestic firms
  - Competitiveness is affected by
    - Productivity — efficient production and technological advancements — reduce firms' costs and allow firms to reduce prices — improve balance of payment — Improve the country's international competitiveness
    - Inflation
      - High inflation — domestic goods and services more expensive relative to foreign goods — deficit — reducing exports and increasing imports
    - The exchange rate
      - Depreciation — makes exports cheaper and imports more expensive — vice versa — lead to an improvement in the balance of payments — improve competitiveness
      - help to restore the effects of high inflation
    - Non-price factors — high quality, design, reputation and reliability
    - Domestic economic activity
      - Rising household incomes — increase in the consumption of home-produced and foreign goods — deterioration in BOP — MPM — buy more raw materials and components from abroad
      - if economic growth has been caused by an improvement in competitiveness and increased exports — improve BOP
    - Global economic activity — demand more exports — rading partners experience strong growth
      - improving the current account
      - size of exports depend on MPM of trade partner
  - Primary and secondary income flows affect the current account of the balance of payments — changes in foreign aid, foreign investment

8.5 Possible conflicts between macroeconomics policy objectives

- Economic growth may affect the environment, the distribution of income, unemployment and the budget balance
  - Economic growth and the environment
    - increases production and consumption, leading to the depletion of natural resources and environmental damage — negative externalities
    - Governments aiming to boost GDP may overlook environmental concerns
    - However, as societies get richer, they can afford to devote more resources to caring for the environment — advanced technology to reduce damage to the environment
  - Economic growth and the distribution of income
    - on average, people get richer
    - benefits are not evenly distributed
      - older industries are often replaced by new one — structural unemployment increases — increases income inequality
      - High-skilled workers and entrepreneurs benefit more from growth — low-skilled workers may struggle
    - does not always lead to greater inequality
      - provide opportunities
      - spend more on merit goods
      - provide cash welfare benefits
  - Economic growth and unemployment
    - compatible
      - If labour productivity increases faster than output — fewer workers may be needed
    - Changing economy — frictional and structural unemployment
    - amount of unemployment depends on
      - amount of change taking place in industries
      - mobility of workers
  - Economic growth and the budget balance
    - compatible
      - economic growth — increase in VAT, indirect tax, direct tax
      - not have to spend as much money on welfare benefits — reduce governemnt spending
    - However, to increase the rate of economic growth, the government may choose to increase its spending, causing budget deficit
      - conflict at least in short-run
      - 描清楚因果关系，谨导致了谁
- How output gaps relate to unemployment and inflationary pressures
  - How a negative output gap relates to unemployment and inflation
    - Aggregate demand is not high enough — involuntary unemployment — demand-deficit unemployment, frictional and structural unemployment
    - there is some spare capacity — moderate inflationary pressures
  - How a positive output gap relates to unemployment and inflation
    - excess aggregate demand — demand-pull inflationary pressures — 8.3 demand-pull inflation
    - not be any demand-deficient unemployment — structural and frictional unemployment also exists
- Use of economic policies to reconcile conflicts between policy objectives
  - Economic growth, unemployment and inflation
    - using monetary and fiscal measures to increase aggregate demand and reduce a negative output gap
      - reduce cyclical unemployment — inflationary pressures unless the productive capacity of the economy increases — go to recession and unemployment will rise
    - supply-side policies — long-run economic growth — overcome short-run conflict between inflation and reducing unemployment and achieving economic growth
      - 

        Supply-side policies can help to reduce the conflict between inflation and unemployment
        - shifting the LRAS curve to the right, shift LRAS1 to LRAS2 — economy to grow without inflation
        - increased real national income from YN1 to YN2 — price level has remained at PL1
  - Economic growth and the balance of payments
    - rising real national income and an increase in aggregate demand
    - supply-side policies and changes in exchange rate can restore competitiveness — long run deficit — accompanied by a high rate of inflation — deteriorate BOP — increase spending on imports
  - Use of supply-side policies to reconcile conflicts between policy objectives
    - resolve conflicts by increasing productive capacity — improving productivity and efficiency and reducing cost of production
    - are not always successful — Governments do not always know what is best for the economy

8.6 International trade

- importance of international trade for an economy
  - International trade: the exchange of goods and services between countries
    - provides consumers with a greater variety of products
      - allows countries to specialise in producing those goods and services they are best at producing — import those products they cannot produce efficiently — better allocation of the world's scarce resources
      - competition that results from the growth in world trade — encourage firms to increase productivity — reduce prices and improve their products
    - does not benefit everyone
      - occupationally and geographically immobile — structurally unemployed for a long time
      - imports are greater than exports — net trade (X – M) will reduce aggregate demand